STATEMENT OF WORK (SOW)

Building 4550 - Elevator Controller Upgrades
US Army Garrison Fort George G. Meade, Fort Meade, Maryland 20755

Work order: FE-00131-4-J
Project Manager: Directorate of Public Works (DPW)
Prepared by: Dipak J. Patel, P.E. (301) 677-9445, Email: dipak.j.patel.civ@army.mil
Date: 08/13/2025

1. Background:

Building 4550 is located at Parade Field Lane, Fort Meade, MD 20755. The building is 4 stories plus a basement. The elevator controls are outdated and need to be upgraded to keep the elevator in service, which is the purpose of this project.

1.1. Site History:

There is one elevator in Building 4550 and the controls are outdated for it. The resident elevator maintenance contractor has indicated that they need to be updated.

1.2. Existing Elevator Manufacture & specification:

Ft. Meade-Bldg. Number	Usage	Manufacture	Drive Type	Stops/Floors	Quantity
4550	Passenger	ETI	Hydraulic	5	1

2. Objective:

This contract is to obtain services to repair and return to service the elevator at building 4550. The Government shall not exercise any supervision or control over the contract service providers performing the services herein. Such contract service providers shall be accountable solely to the Contractor who, in turn is responsible to the Government.

3. General Requirements:

3.1. Scope of Work:

Provide a proposal which will be the modernization of one (1) passenger elevator at Building 4550 which is serving from the basement to the fourth (4^{th}) floor.

Elevator modernization encapsulates the process of upgrading essential mechanical systems, controls, and safety features of an elevator to align with contemporary latest code & standards. It may also encompass aesthetics of the interior if budgets allow. This vertical transportation modernization is crucial for improving efficiency, ensuring compliance with the latest safety regulations, and significantly enhancing the user experience. An elevator modernization starts a new life cycle of elevators and plays a vital role in maintaining the building's competitiveness and value.

Elevator modernization involves updating the critical mechanical components of an elevator system to improve its efficiency, safety, and performance. The core of modernization lies in updating the existing elevator system, making them more reliable, enhancing performance and aligning them with current technological and safety standards. The proper elevator system must be installed to ensure the desired life cycle of 10 – 15 years is captured. A building must follow the best practice methodology to ensure all appropriate components are reused, recycled, refurbished or replaced.

Modernize the subjected elevator at Bldg. 4550 located on Ft. Meade, Maryland. The unit in question was modernized in the early 2000s and the controls (elevator controller, door equipment, fixtures, pumping unit) are beyond their respective life expectancy and have been showing details of wear and tear for many years to date. Plan on removing all the following equipment and installing the latest state of art American Made Non-Proprietary equipment that far durable, reliable and dependable than any other within our industry.

1. Passenger Elevator #1:
    a. New Elevator Submersible Pumping Unit
    b. New Controller
    c. New Elevator Signals/Fixtures
    d. New Hoist way Doors
    e. New Hoist way Door Equipment
    f. New Car Door Equipment
    g. New Car Door
    h. New Elevator Wiring to include Multicable, Signal Wiring and Traveling Cable
    i. New Cab Interior

Elevator submersible pump replacement scope of work: The following scope of work for replacing an elevator submersible pump, covering aspects of replacement, potential issues, and relevant regulations.
SUBMERSIBLE PUMPING UNIT (BOREMAX or EQUIVALENT) assembled unit consisting of positive displacement pump, induction motor, master-type control valves combining safety features, holding, direction, bypass, stopping, manual lowering functions, shut off valve, oil reservoir with protected vet opening, oil level gauge, outlet strainer, drip pan, all mounted on isolating pads. Provide thermal units to maintain oil at operating temperature. Enclosed the entire unit with removable sheet steel panels lined with sound absorbing material. Provide closed transition SCR soft start. Design unit for 80 upstarts per hour.

1. preparatory work

- Safety:
    - Disconnect power to the pump prior to initiating any repair work.
    - Ensure the work area is clear of obstacles and potential hazards.
    - Utilize appropriate personal protective equipment (PPE).
- Sump Pit: If necessary, clear debris from around and within the sump pit to prevent contamination of the well.

2. Removal of existing pump

- Disconnection:
    - Unplug the old submersible pump.
    - Loosen the T-bolt(s) on the check valve to disengage the piping connected to the pump, being careful not to lose the bolt into the pit.

    - Remove the old submersible pump from the sump pit.

- Cleaning: Use a shop vac or other suitable method to clean any debris from the pit, as sump pumps collect groundwater.

3. Installation of new pump

- Pump Selection: Ensure the new submersible pump has the appropriate horsepower and discharge pipe size to match the existing piping and operational requirements.
- Assembly: Attach the discharge pipe to the new pump, potentially reusing sections of the old pipe if compatible.
- Sump Pit Placement: Lower the new pump into the pit, ensuring it sits flat.
- Float Valve: Adjust the float valve's wire or tubing to prevent it from getting stuck against the side of the pit.
- Check Valve and Piping: Reconnect the check valve and tighten all connections and fittings.

4. Post-installation and testing

- Power Restoration: Plug the pump back in.
- Testing:
    - Pour water into the pit to verify the pump's proper functioning and float activation.
    - Check for a good flow of water from the discharge pipe.
- Alarm Systems: Verify the functionality of the oil sensing control system and its alarms.
- Pumping Tests: For specific well or aquifer assessments, conduct pumping tests at various flow rates to determine performance characteristics and hydraulic parameters.
- Documentation: Record the installation date on the pump.

5. Compliance and codes

- Safety Codes: Adhere to all applicable safety regulations and codes, such as OSHA and the NEC, throughout the installation process.
- Elevator Sump Pump Requirements: Ensure the sump pump and its installation comply with relevant elevator sump pump codes, which may include requirements for independent piping, discharge capacity, oil sensing control systems, and alarms.
- Prohibited Locations: Avoid locating plumbing systems, including elevator sump pumps, in elevator shafts or equipment rooms, with the exception of floor drains, sumps, and sump pumps at the base of the shaft if indirectly connected to the plumbing system and compliant with Section 1003.4 of the International Plumbing Code.

6. Considerations

- Elevator Downtime: Plan the replacement to minimize elevator downtime, as the elevator will be out of service during the pump replacement.
- Difficult Installations: Deep or hard-to-access wells can increase the time required for pump replacement.
- Maintenance: Regular maintenance and inspections of the elevator sump pump are essential to prolonging its lifespan and identifying potential issues early on.

B. Elevator new controller replacement statement of work (SOW): An elevator controller replacement involves several following key steps:

NEW CONTROLLER (MCE or EQUIVALENT), Magnet operated with contacts of design and material to ensure maximum conductivity, long life and reliable operation without overheating or excessive wear. Provide wiping action and means to prevent sticking due to fusion. Contacts carrying high inductive currents will be provided with deflectors or suppressors. 3. Microprocessor-Related Hardware: a. Provide built-in noise suppression devices which provide a high level of noise immunity on all solid-state hardware and devices. b. Provide power supplies with noise suppression devices. c. Isolate inputs from external devices (such as push buttons) with opt isolation modules. d. Design control circuits so that one side of the power supply is grounded. e. Safety circuits shall not be affected by accidental grounding of any part of the system. f. System shall automatically restart when power is restored. g. System memory shall be retained in the event of power failure or disturbance. h. Equipment shall operate properly with a 500 KHZ to 1300 MHZ radio frequency signal, transmitted at a power level of not less than 100 watts of Effective Radiated Power (ERP) at 3 feet. i. Equipment will be provided with Electro Magnetic Interference (EMI) shielding within FCC guidelines. 4. Wiring: CSA labeled copper for factory wiring. Neatly rout all wiring interconnections and securely attach wiring connections to studs or terminals. 5. Permanently marked components (relays, fuses, PC board, etc.) with symbols shown on wiring diagrams. 6. Provide reduced voltage motor starting circuits with a solid-state motor starter. E. Piping and Oil: Provide new piping connections and oil for the system. Provide isolated pipe stands or hangers as required.

1. Planning and design

- Based on the assessment, designing a customized upgrade plan, including selecting the appropriate controller and planning the installation process.
- Ensuring compliance with all relevant safety regulations and codes, such as ASME A17.1 and ADA requirements.
- The controller is the "brains" of the elevator system, and replacing it is often a major component of an elevator upgrade.

2. Installation

- Qualified technicians perform the installation of the new controller, minimizing downtime and disruption to the building's operations.

3. Testing and calibration

- After installation, thorough testing and calibration are conducted to ensure the new controller functions optimally, including checking safety features, response times, and energy efficiency.
- The ASME A17.1/CSA B44 standards and IBC 2021 code updates include stricter requirements for emergency communication systems, including video communication systems and two-way communication with both visual text and audible features.

4. Ongoing maintenance

- Establishing a plan for ongoing maintenance to ensure the long-term reliability and performance of the upgraded system.
- Provides ongoing elevator inspection, maintenance, testing, repairs, parts replacements, and modernizations under federal contracts.

- The contractor is typically responsible for providing labor, parts (new and genuine, equal to OEM parts), tools, cleaning supplies, and lubricants required for the services.

C. Elevator hoist way doors SOW:
HOISTWAY DOORS (10) (GUNDERLIN or EQUIVALENT) Provide new elevator hoist way entrance doors reusing the present frames and saddles (sills). 2. Each door panel will be hollow metal, 1-1/2-hour test construction, manufactured of cold rolled furniture steel, flush design both sides, rigidly reinforced, sound deadened and will bear approved label.

- Car Doors (2) and Hoist way Doors to be #4 Stainless Steel

HOISTWAY DOOR EQUIPMENT, Provide a sheave type two-point suspension hanger and track for each hoist way door. Sheaves shall be hardened steel, not less than 3 1/4 inches in diameter with sealed grease-packed precision ball bearing. 2. The upthrust shall be taken by a roller mounted on the hanger and arranged to ride on the underside of the track. 3. The track shall be of cold rolled steel or cold drawn steel and shall be rounded on the track surface to receive the hanger sheaves. The track shall be removable and shall not be integral with the header.

(GAL or EQUIVALENT) DOOR OPERATORS (4) FOUR FRONT OPENINGS AND (1) ONE REAR OPENINGS Car Door

Hangers, Sheaves, Tracks and Gate Switch (New) 1. Provide a sheave type two-point suspension hanger and track for each car door. Sheaves shall be hardened steel, not less than 3 1/4 inches in diameter with sealed grease packed precision ball bearing. 2. The upthrust shall be taken by a roller mounted on the hanger and arranged to ride on the underside of the track. 3. The track shall be of cold rolled steel or cold drawn steel and shall be rounded on the track surface to receive the hanger sheaves. The track shall be removable and shall not be integral with the header. 4. Provide a new gate switch that connects directly to the car door track. The gate switch shall prevent movement of the elevator until such time as it signals the control equipment that the car door has physically closed.

H. Car Doors (New) 1. Provide standard 1” think, 14-gauge hollow metal flush construction panels, reinforced for power operation and insulated for sound deadening. 2. The panels shall have no binder angles and welds shall be continuous, ground smooth and invisible. 3. Drill and reinforce panels for installation of door operator hardware, door protection device, door gibs, etc. a. Provide each door panel with two removable laminated plastic composition guides arranged to operate in the sill grooves with minimum clearance.

- Regarding hoist way doors, these are doors at each floor landing that open into the elevator shaft. They are also referred to as landing doors.
- Modernization projects often involve upgrading various components of the elevator system, including hoist way doors. Modernizing hoist way doors can improve their safety, reliability, and functionality. For example, in one modernization project, the scope of work included installing new stainless-steel elevator hall doors at each landing.
- Safety regulations are a crucial aspect of elevator operations and modernization. According to ASME A17.1 standards, the space between swinging hoist way doors and folding car doors must reject a 4-inch-diameter sphere at all points when both doors are closed.
- Installation of hoist way doors can be done using the moving elevator car platform. Proper alignment is essential and can involve using tools like plumb lines to verify the distance from the plumb line to each jamb.

D. Elevator flush-mounted hall fixtures and position indicators:

FLUSH MOUNTED HALL FIXTURES AND POSITION INDICATORS (INNOVATION or EQUIVALENT): Provide new corridor pushbuttons as further described: a. A riser of new surface-mounted security key switch & pushbutton signal fixtures will be provided on each landing. Each new signal fixture will consist of a faceplate and illuminating pushbuttons measuring 3/4” at their smallest dimension as selected by the Owner, a recessed mounting box, electrical conduit, and wiring. New key switch activated pushbutton signal fixtures will be mounted directly over the existing corridor stations at a centerline height of 42” above the floor. The new signal fixture faceplate will be installed both plumbs to the finished wall. b. Each intermediate landing will be provided with signal fixtures containing two (2) pushbuttons while terminal landings will be provided with fixtures containing a single pushbutton. These buttons will be arranged to become illuminated upon use of the key switch on an individual basis as “up” or “down” calls are registered and will remain so until canceled. c. Existing corridor key switch fixture cover plates will be removed along with all related electrical conduit and wiring. d. Provide engraving of the fixture cover plates as required by AHJ requirements. e. Provide the required engraving of the main level fixture cover plate to include fireman’s service operation key switches, fireman’s hat jewel, etc., ASME A17.1 requirements.

MAIN OPERATING PANEL, Provide a main car operating push button panel on the inside front return panel of the car. 2. Pushbuttons will be provided for each floor served and will cause the car to travel to the floor on momentary pressure of the button. 3. The pushbuttons will become individually illuminated as they are pressed. The button lights will be extinguished as the calls are answered. LED-type bulbs

1. Scope of work

- Supply: Provide elevator flush-mounted hall fixtures, including pushbuttons (available in various finishes like stainless steel and Muntz, and with options for digital, Lexan, or custom floor indicators).
- Supply: Provide elevator flush-mounted hall position indicators.
- Supply: Include digital position indicators, custom engraved logos, fire evacuation pictographs, and engraved floor maps as requested.
- Installation: Install the fixtures and indicators according to manufacturer instructions, ensuring ADA code compliance, including a minimum height of 72 inches above the finished floor for indicators, and arrows at least 2.5 inches tall that are visible from the call buttons.
- Wiring: Connect the fixtures and indicators to the elevator control system as per provided wiring diagrams.
- Testing and Commissioning: Verify proper operation of the fixtures and indicators after installation.

2. Features and benefits

- Flush-mounted design: Integrates seamlessly into the building architecture.
- Durable materials: Built to withstand heavy use and potential vandalism.
- ADA compliant: Ensures accessibility for all users.
- Clear and visible indicators: Provides easy identification of floor level and travel direction.
- Customization options: Allows for tailored designs and aesthetic integration.

3. Maintenance

- Refer to the manufacturer's recommendations and specifications for proper maintenance of the fixtures and indicators.
- Consider utilizing the services of a qualified elevator maintenance provider for routine inspections and upkeep.

  Main car operating panel

E. Car operating panel (COP)

- This is the panel inside the elevator car that passengers use to select floors and control other functions.
- It typically includes floor buttons, door control buttons, and emergency buttons, along with a display panel to show the current floor and direction of travel.
- Key switches might be present for building personnel and emergency services.
- Modern COPs can also incorporate features like video displays for information and advertising.
- In this context, it would specify the tasks and requirements for working with the elevator's main COP, which could include:
    - Installation: Installing a new COP or integrating it into an existing elevator system.
    - Modernization: Upgrading the COP with new features or technologies, like media screens or access control systems.
    - Maintenance or Repair: Addressing issues or ensuring the proper functioning of the COP.
    - Integration with other systems: Connecting the COP to access control systems, building management systems, or other external systems.

F. Provides communication and emergency signaling devices for elevators, adhering to safety codes like ASME A17.1/CSA B44 and ADA guidelines. These devices ensure passengers can communicate with emergency services or designated personnel during an emergency, such as an entrapment or medical event.

COMMUNICATIONS AND EMERGENCY SIGNALING DEVICES A. Telephone (New) 1. Provide an automatic dialing, hands-free telephone in the new car station without a separate faceplate. All components will be mounted to the back of the panel. 2. The system will be arranged to place a call to a central point as selected by the Owner. Provide an automatic shut-off feature and a push button to initiate a call.

WIRING: Provide all new wirings to include a new travel cable, multi-wire and other necessary conductors to make operations of the elevator, and its equipment work as intended.
Key requirements and features

- Two-way communication: Systems facilitate two-way communication (both voice and visual) between elevator occupants and a monitoring center or authorized personnel.
- Video and text messaging: The systems are designed to allow communication via video and text messages, a crucial feature for hearing and/or speech-impaired passengers, and a requirement in the updated IBC 2021 code.

- Accessibility (ADA compliance): Emergency phones and communication devices must be accessible to individuals with disabilities, including tactile buttons no higher than 48 inches from the floor, along with visual and audible signals.
- 24/7 monitoring and availability: The system must be monitored 24/7, by trained emergency responders or designated personnel, and cannot be turned off, even after hours.
- Automatic location identification: The system should automatically identify and transmit the exact location of the elevator to emergency responders.
- Reliability and backup power: The system must function for at least four hours during a power outage and should have a backup power source, such as a cellular or VoIP line, for increased reliability.
- Installation considerations: Installation may involve cellular or VoIP lines as alternatives to traditional POTS lines, depending on the building's infrastructure and local fire codes.
- Regular testing: Regular testing is mandated to ensure that emergency phones are functioning properly and comply with safety codes. This includes activating multiple phones simultaneously to test two-way communication and verifying that the system can automatically identify the elevator's location.

MISC: F/A contractor to integrate new elevator controls with existing F/A modules to activate Phase I, II and any other shunt trip activation if applicable.

- New Hydraulic Oil
- New Hydraulic Line
- New Hoist way Safety Switches (Limits, Pit Stop, Cartop Box)
- New Electrical Conduit, Duct and Raceway
- New Interior Elevator Returns to be clad

CAB INTERIOR (SNAP-CAB or EQUIVALENT): Provide a new cab shell based upon the selection of the customer.

Reuse / Retain:

Hoist way Entrances (Reuse)
Retain Existing Guide Rails (Reuse)
Retain Existing Buffers: Spring type with blocking and supports.
Frame: Retain Existing Retain Existing Platform
Retain Jack: Currently, Jack Assembly is in good working order.

The contractor shall determine if all the above scope is required or parts thereof prior to starting work and give a detailed analysis of what is needed and a corresponding cost of same.

Prior to commencing work, all appropriate professionals shall visit the site to examine the conditions under which the work is to be performed to identify those conditions that might adversely affect the work. These conditions that may adversely affect work may include, but may not be limited to the following:

Observation, Analysis, and Evaluation of existing conditions, dimensions, and physical characteristics to validate constructability and the proposed installation (per this Statement of Work).

Incidental work related to the return to service of the elevator in Bldg. 4550 (Work sequencing, current materials, proposed finishes, utilities, etc.)

Field verifies conditions, elevations, and dimensions prior to initiating work.

Contractor shall notify the Contracting Officer of any discrepancies, interferences, or contradictions between contract documentation and field conditions by submitting a request for clarification.

Affected phases of work shall not be initiated until all requests for clarification have been answered.

Work with COR to gather requirements, verify requirements, and establish metrics.

Develop site pre-construction and construction submittals for COR review and approval.

The contractor shall create and present an adequate project plan including a Work Breakdown Structure w/ timeline i.e. Activity Schedule, Finish Schedule (material and appearance selection listing), and Handoff/Closeout plan.

Most of the work associated elevator shall be accomplished from late Friday after 15:00 to early Monday morning with restoring back to the business to support weekday operation or plan to arrange work around long weekend hours to minimize elevator downtime during weekday or approved by COR with advance notice.

All work shall be coordinated with the facility POC for this project which has been Identified as: C. Wes Smith, 240-328-4103c Charles.w.smith89.civ@army.mil , and the DPW COR for the project to be determined. Especially any scheduled outages of the elevator in question for the work to be completed.

The contractor shall present project plans to the government client for review and approval.

3.1.1. Brief Work Description:
Upgrade controller and operating system for the elevator in Bldg. 4550

3.1.2. Specific Tasks:
This proposed project includes, but is not limited to:

3.1.2.1. Design/Prep Phase Tasks:

3.1.2.1.1.
The Contractor shall conduct a site visit survey. To ensure proper equipment and materials are needed to complete the task at hand. The contractor shall ensure the project meets or exceeds Federal, State and/or Local Code Regulations. The survey shall include but not limited to elevation changes, electrical needs, equipment needs etc.

3.1.2.2. Demolition Tasks:
Contractor shall perform the following, but not limited to the following to meet industry standards:
1) The contractor shall be responsible for removal of all excavated and excess material accumulated during this project.
2) Contractor shall disconnect all components of unit controller supply wiring that exist. The wiring shall be inspected and removed if in inadequate condition or inadequate size for the new controller and system upgrades.
3) The contractor shall disconnect, remove and dispose of all components of the old unit controller and operating system parts that are to be replaced.

3.1.2.3. Installation Tasks:
Contractor shall perform the following, but not limited to the following to meet industry standards:
1) Contractor shall install a new controller and operating system parts for the elevator at Bldg. 4550.

3.1.2.4. Road Closures:
The contractor will ensure that all road closure requests are submitted at least two (2) weeks in advance of when the closure is required.

This request should include at a minimum the location & anticipated duration of closure.

Requests shall be properly permitted through the State, County, or City prior to road closure.

This request must be reviewed by the COR, Provost Marshals Office (PMO), Fire Department, and approved by the Garrison Commander before any road closure will be allowed to take place. It is recommended that any road closure request be submitted as soon as possible to avoid delays.

3.1.3. Testing:
- Contractor shall test and confirm in writing to the managing COR "Contracting Officer Representative" that the new elevator controllers and operating system are one hundred percent operational and functional upon final installation under this contract. Contractor shall provide a one hundred percent turnkey fully operational elevator that meets any applicable standards in:
- EM 385-1-1 U.S. Army Corps of Engineers Safety and Health Requirements Manual
- UFC 1-200-01 DoD Building Code (General Building Requirements)
- COR will review approval and accept the systems if the systems are found acceptable.

3.1.4. Final Inspection:
The Contracting Officer or COR will conduct the final inspection of the project; with at least the Contractor and Building Tenant. The end user requires a turnkey product that operates as intended.

3.1.4.1.
Listed deficiencies on the punch list should have been corrected at this point. However, deficiencies that cannot be corrected due to operation or seasonal weather can be rectified on dates approved by the Contracting Officer. All debris management shall be complete.

3.1.5. Equipment:
The contractor shall provide all ancillary equipment, manpower and support, to design demolish and install components mentioned within this summary of work. The contractor shall be responsible for obtaining all dimensions and measurements of areas and components to be installed prior to installation of components.

3.1.6. Technical:
Where necessary, the Contractor will be required to coordinate and cooperate with Fort Meade, Directorate of Public Works, for access, data and information sharing.

The requirements contained herein are the minimum necessary and should be supplemented by the Contractors own design and quality control requirements as may be applicable. The Contractor shall coordinate, demolish, design, develop, fabricate, assemble, and install new components as requested in the SOW and drawings provided for this contract.

The Contractor shall field verify conditions, elevations and dimensions prior to initiating work. The Contractor shall notify the Contracting Officer of any discrepancies, interferences or contradictions between contract documentation and field conditions by submitting a request for clarification. Affected phases of work shall not be initiated until all requests for clarification have been answered.

3.1.6.1. Utilities marking:
When applicable, obtain digging permits prior to the start of excavation, and comply with Installation requirements for locating and marking underground utilities. Contact local utility locating service (MISS Utility) a minimum of 72 hours prior to excavating, to mark utilities, and within sufficient time required if work occurs on a Monday or after a Holiday. Verify existing utility locations indicated on contract drawings, within area of work. Provide the COR with tracking documentation.

The contractor shall be responsible for identifying and marking any/all utilities within the affected area(s). Marking and identifying will be 100% solely the contractor’s responsibility and not government personnel. Scan the construction site with Ground Penetrating Radar (GPR), electromagnetic, or sonic equipment, and mark the surface of the ground or paved surface where existing underground utilities are discovered. Verify the elevations of existing piping, utilities, and any type of underground obstruction may reside. Verify elevations before installing new work closer than the nearest manhole or other structure at which an adjustment in grade can be made.

3.1.7. Existing Work:
In addition to FAR 52.236-9, Protection of Existing Vegetation, Structures, Equipment, Utilities, and Improvements:

3.1.7.1. Removal:
Remove or alter existing work in such a manner as to prevent injury or damage to any portions of the existing work which remain.

3.1.7.2. Repair and Replace:
Repair or replace portions of existing work which have been altered during construction operations to match existing or adjoining work, as approved by the Contracting Officer. At the completion of

operations, existing work must be in a condition equal to or better than that which existed before new work started.

3.1.7. Codes:
The Contractor shall coordinate and obtain access to the project site. The Contractor shall coordinate and obtain any permits to do the work requested herein.

Contractors shall abide by all State, Federal and local construction regulations and/or codes for the duration of this project regarding construction and safety procedures and methods used while performing this project.

All materials shall be installed in accordance with the approved recommendations of the manufacturer as well as conform to the contract documents. The installation shall be neat in appearance, free from defects and shall be accomplished by workman skilled in this type of work.

3.1.7.1. Occupied Facility:
When applicable the Contractor shall ensure construction activity noise levels do not exceed 80 decibels.

When performing work in occupied buildings, the facility and contents shall be always kept secure. Provide temporary closures as required to maintain security as directed by the Contracting Officer.

During the construction, the area surrounding the construction zone may be occupied with building personnel. The contractor is required to prevent the spread of any debris and dust. The contractor shall isolate the construction zone by hanging 6 mil thick polyethylene sheeting from the ceiling to floor when applicable. All openings shall be sealed with a tape that is strong enough to last during the duration of the construction. The access opening for the contractor shall be overlapped to prevent dust from escaping the construction area.

3.1.8. Clean Up:
The contractor shall be responsible for the daily clean up and disposal of any debris with the performance of this project. Work areas should remain clean and safe to the furthest extent possible to reduce safety concerns. It is the contractor's responsibility to ensure all debris is disposed of at an off-post landfill in accordance with all local, state and federal regulations.

3.1.9. Superintendent:

3.1.9.1. Minimum Communication Requirements:
Have at least one qualified superintendent, or competent alternate, capable of reading, writing, and conversing fluently in the English language, on the jobsite always during the performance of contract work. In addition, if a Quality Control (QC) representative is required on the contract, then that individual must also have fluent English communication skills.

3.1.9.2. Superintendent Qualifications:
The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance. The individual must be capable of interpreting a critical path schedule and construction drawings. The qualification requirements for the alternate superintendent are the same as for the project superintendent. The Contracting Officer may

request proof of the superintendent's qualifications at any point in the project if the performance of the superintendent is in question.

3.1.10. Delivery Schedule:
Notify the Contracting Officer by writing at least 14 calendar days in advance of the date on which the materials and equipment are required. Pick up materials and equipment no later than 30 calendar days after such date.

Materials and equipment will be available on or after 14 calendar days after the award of contract.

3.1.11. As-builts:
At the time of the final inspection or earlier the contractor shall provide As-Built drawings to the Contracting Officer, and the Contracting Officer's Representative. Contractor shall provide One (1) AUTOCAD DWG. Drawing of the water line connection and one PDF, electronically, upon completion of the project to the Federal Government, preferably via e-mail to the managing COR. Federal Government will provide an AUTOCAD drawing to the contractor to make this as built so.

3.1.12. Warranty:
The Contractor shall warranty all work for a minimum of one (1) year after Government acceptance per FAR 52.246-21. Under this warranty, the Contractor shall remedy any nonconformities or defects at the Contractor's expense. The Contractor shall also identify, complete and deliver manufacturer warranty documents for any materials with manufacturer warranties longer that one (1) year.

3.2. Period of Performance:
The Contractor shall complete work within 180 days after KO issues the Notice to Proceed (NTP).

3.3. Place of Performance:
The place of performance shall be at B4550 Parade LN, Fort Meade, MD.

3.4. Technical Point of Contact:
Dipak J. Patel, P.E., Fort Meade, MD 301-677-9445
Donnell Flemming, Fort Meade, MD 301-677-1771.

3.5. Hours of Operation:
The hours of operation for this project are between the hours of 7:00 am and 5:00 pm. If weekend work is required for project completion alternate work hours can be coordinated with the COR.

3.6. Project Tracking:
Prepare for approval a Project Schedule, as specified herein, pursuant to FAR Clause 52.236-15 Schedules for Construction Contracts. The contractor shall provide project submittals that give a clear overview of project progress.

3.6.1 Project Schedule:
The project schedule shall be provided to the COR within 10 working days of the award and should be submitted in Microsoft Projects, or an approved equal format.

3.6.2. Schedule of Values:
Payments will not be made until the Schedule of Values has been submitted to and accepted by the Contracting Officer

The schedule of values shall show a minimum of the cost delineation for the materials and construction divisions.

3.7. Safety:
The number for EMS&FIRE is 911.

The Contractors must arrange for their own emergency medical treatment in accordance with EM 385-1-1. Government has no responsibility to provide emergency medical treatment.

The Contractor is solely liable for ensuring safe work and otherwise safe actions, behavior, and conduct while on site. It is understood the Contractor is solely responsible for following all applicable Federal and State of Maryland mandates and guidelines. The contractor's Safety Plan will be reviewed by base safety which will take 2 business weeks.

In addition to the detailed requirements included in the provisions of this contract, comply with the most recent edition of USACE EM 385-1-1, and the following federal, state, and local laws, ordinances, criteria, rules and regulations. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

3.7.1.
All safety violations identified by the Fort Meade Safety Office shall be reported to the COR for the Contractor Superintendent's disposition. In cases of life safety, the Fort Meade Safety Office may act independently to mitigate the safety hazard and report to the COR when practical. In all cases of life safety issues, the Contractor Superintendent must report actions taken by the Contractor to the COR within one (1) hour; the COR must subsequently report the issue to the Department of Public Works Director, IAW critical incident report procedures and the Fort Meade Safety Office. In all other cases, the Contractor Superintendent must report the safety issue to the Fort Meade Safety Office within eight (8) hours of the incident. Initial reporting may be done telephonically. Subsequently, all reports must be provided in writing within a time to be determined by the Fort Meade Safety Office IAW Army regulations.

3.7.2. Health and Safety:
The Contractor shall be fully responsible for and shall assume all liability for compliance with all applicable regulations pertaining to the health and safety of personnel during execution of work. The Contractor shall hold the Government harmless for any action on its part or that of its employees, subcontractors or Contractor's representatives, which results in illness, injury, or death to anyone on the Fort Meade grounds. All rules of safety which are or may be imposed upon the Contractor by federal, state or municipal code, and the applicable regulations shall be effectively carried out in the performance of this contract.

3.7.3. Health and Safety Plan:
The contractor shall complete an Accident Prevention Plan and a Hazard Risk Analysis for all work to be performed on the project. A qualified person must prepare the written site-specific (AAP).

3.7.4. Fire Safety:
Provide product data for review for any fire stopping products to be used. All penetrations made in fire rated assemblies shall be filled with the proper fire rated material that either meets or exceeds the current rating of that assembly. All penetrations made and subsequently filled with the appropriate fire stopping shall be inspected by a Ft. Meade Fire & Emergency Services, Fire Prevention Division representative, before concealing, painting, etc.

If any Hot Work Operations are to be performed, i.e. Torch applied membranes, Tar pot, Soldering, welding, or cutting, then a Hot Work Permit shall be required prior to the commencement of work. To obtain a Hot Work Permit contact the Fire Department at 301-619-2528.

3.8. Security Requirements:
Contractor personnel or any representative of the Contractor entering Fort Meade shall abide by all security rules and regulations. The Contractor shall carry passes issued by Fort Meade for all vehicles entering the base, whether company-owned or employees' private vehicles.

3.8.1. Physical Security:
The contractor shall be responsible for safeguarding all government equipment, information and property provided for contractor use. At the close of each work period, government facilities, equipment, and materials shall be secured.

3.8.2. Gate Access Requirements:
IAW Army Directive 2014-05 all personnel desiring unescorted access to Fort Meade will enter the installation through an authorized Access Control Point (ACP) and be vetted using the National Crime Information Center (NCIC) interstate Identification Index (III). The Installation Commander will, in the absence of a waiver deny uncleared contractor, subcontractor and visitors unescorted access to the installation based on the results of the NCIC-III check that contains credible derogatory information. Such derogatory information includes, but is not limited to the following:

3.8.2.1.
NCIC III contains criminal arrest information about the individual that causes the installation Commander to determine that individual presents a potential threat to the good order, discipline, or health and safety on the installation.

3.8.2.2.
The Installation is unable to verify the individual's claimed identity in the attempt to gain access.

3.8.2.3.
The individual has a current arrest warrant in NCIC, regardless of the offense or violation.

3.8.2.4.
The individual is currently barred from entry or access to a federal installation or facility.

3.8.2.5.
The individual has been convicted of crimes encompassing sexual assault, armed robbery, rape, child molestation, production or possession of child pornography trafficking in humans, drug possession with the intent to sell or drug distribution.

3.8.2.6.
The individual has a US conviction for espionage, sabotage, treason, terrorism or murder.

3.8.2.7.
The individual is a registered sex offender.

3.8.2.8.
The individual has a felony conviction within the past 10 years, regardless of the offense or violation.

3.8.2.9.
The individual has been convicted of a felony firearms or explosives violation.

3.8.2.10.
The individual has engaged in acts or activities designed to overthrow the U.S. Government by force.

3.8.2.11.
The individual is identified in the Terrorist Screening Database (TSDB) as known to be or suspected of being a terrorist or belonging to an organization with known links to terrorism or support of terrorist activity.

3.8.2.12.
Individuals are barred from Fort Meade.

3.8.3. Access Denial Waiver Process:
In cases where an unclear contractor, subcontractor is denied access based on derogatory information obtained from an NCIC or NCIC III check, the Installation Commander shall offer an Access Denial Wavier Application packet only if the individual requests a waiver.

3.8.4. Anti-terrorism Security Requirements:
The contractor and all associated subcontractors' employees shall comply with applicable installation, facility, and area commander installation and facility access and local security policies and procedures (provided by the Government COR). The contractor should also provide all information required for background checks to meet installation access requirements to be accomplished by the installation Directorate of Operations Protection Branch, Visitor Center and Security Office. The contractor workforce must comply with all personal identity verification requirements as directed by DoD, HQDA, and/or local policy. In addition to the changes otherwise authorized by the changes clause of this contract, should the Force Protection Condition (FPCON) at any individual facility or installation change, the Government may require changes in contractor security matters or processes. The COR is responsible for making sure items selected on this sheet are complied with. All access control requirements are to be adhered to. All contractors and sub-contractors need to be vetted through the visitor center, and obtain visitors passes to work on post. COR needs to conduct spot checks at work sites to make sure contractors follow visitor requirements and responsibilities. AT Level 1 and iWatch Awareness training assist in informing

the contractor that at some times Random Antiterrorism Measures (RAM) may be in effect, where their vehicle may be checked, ID cards validated, security checks done by COR and Police. Gate operations could be affected based on the FPCON or RAM. COR is responsible, depending on the length of this contract by signing at least one contractor up for safety messages for weather or incident alerts, that POC will be responsible for notifying the other contractors by phone tree or other means. The COR is also responsible for ensuring that any blueprints that show information outside of the scope of work are either collected back at the end of the job or destroyed.

3.8.4.1.

Anti-Terrorism Level (1) One Training. All contractor employees, including subcontractor employees, requiring access to Army installations, facilities, or controlled access areas shall complete AT Level I awareness training within 30 calendar days after contract start date or effective date of incorporation of this requirement into the contract, whichever applies. All contractor employees must complete the annual AT Level 1 awareness training. Annual training is to be conducted in the anniversary month when it was initially completed. The contractor shall submit certificates of completion for each affected contractor employee and subcontractor employee to the COR (or to the contracting officer, if a COR is not assigned) within 15 calendar days after completion of training by all employees and subcontractor personnel. AT Level I training is now conducted on the Joint Knowledge Online (JKO) site. For those that have a CAC, AT Level I can be accessed via https://jkodirect.jten.mil For those personnel that do not have a CAC they can utilize the below instructions to complete, i.e. contractors etc.

1. Google JKO and begin the process from there. http://jko.jten.mil

   Clicks:
2. No DOD CAC.
3. I am a US mil, government civil servant or contract employee.
4. I've been directed to take required training on JKO.
5. Courses.
6. I do not have a .MIL, .GOV, or.NDU.EDU address or I am a Multi-National Student.
7. Fill out the contact sheet and email to sponsor (In my opinion this is the COR) 8. The sponsor will email the JKO help desk. (Again, in my opinion this is the COR) Approval
    a. Help desk sends non-CAC users an email with User ID.
    b. Help desk will send a separate email with an electronic token to register a new password (token is good for 24 hours).
    c. The new User follows the instructions in the email and enters a new password.
    d. Users will need to enroll in Course # JS-US007-14 (User has up to 1 year to complete the course) (Contractors need to follow the requirements outlined in the AT/OPSEC cover sheet and complete it in 30 days).

If training is required in a non-English language, please contact Fort Meade ATO at usarmy.meade.usag.mbx.dptms-antiterrorism@mail.mil.

3.8.4.2.

iWatch Awareness Training. The contractor and all associated subcontractors shall brief all employees on the local iWatch program (training standards provided in this PWS by Fort Meade ATO). This locally developed training will be used to inform employees of the types of behavior to watch for and instruct employees to report on suspicious activity to the COR. This training shall be completed within 30 calendar days of contract award and within 30 calendar days of new employees' commencing performance, with the results reported to the COR no later than 5 calendar days after completion.

3.8.5. Operational Security (OPSEC) Requirements:
Personal information or information considered critical to the installation's operations released intentionally or unintentionally could have direct consequences for military and government personnel as well as our critical operations. Therefore, the contractor shall be responsible for safeguarding personal information and information considered critical to Fort Meade's operations. Per AR 530-1, Operations Security (OPSEC), new contractor employees must complete Annual OPSEC training within 30 calendar days of reporting for duty. All contractor employees must complete annual OPSEC awareness training. All contractor employees shall adhere to USAG Fort Meade OPSEC policies, regulations and guidance. OPSEC awareness training is required when working with personal information (Social Security Number, date of birth, etc.) or on government systems that are considered critical to the installation's operations. Contractor personnel shall not release any personal or operational information (No photography) to the public, press, or any other personnel or organizations without prior approval from the USAG Command.

3.8.5.1. OPSEC Awareness Training:
New contractor employees and subcontractor employees must complete Level I OPSEC awareness training within 30 calendar days of reporting for duty. All contractor employees must complete annual OPSEC awareness training. Annual training is to be conducted in the anniversary month when it was initially completed. The contractor shall submit certificates of completion for each affected contractor employee and subcontractor employee to the COR (or to the contracting officer, if a COR is not assigned) within 15 calendar days after completion of training by all employees and subcontractor personnel.
(http://www.meade.army.mil/dptms/pdf/OPSECTrainingContractors.pdf)

3.8.5.2. OPSEC Standing Operating Procedure:
The contractor shall follow the Fort Meade OPSEC SOP while performing work on Fort Meade.

3.8.5.3. Industrial Personnel Security:
IAW AR 380-67, AR 380-49 and DOD 5200.2-R, all contracts, classified, sensitive but unclassified and unclassified are required to contain a security clause. : IAW Army Regulations 380-67, 380-5 and 380-49, the Contractor shall ensure all personnel performing under this contract have a completed National Agency Check with Local Checks (NACLC) or Unclassified contracts require a NACI and that it contain no unfavorable information and has been completed by Office of Personnel Management and/or adjudicated by Defense Industrial Security Clearance Office (DISCO) prior to the start of the contract. The security requirements must be maintained for the life of the contract, and citizenship will be verified by presenting an original birth certificate or a valid US Passport or the original Naturalization Certificate to the USAG Security Office at the time of processing for all contract personnel assigned to the contract. To maintain the security requirements any derogatory information, including law enforcement issues, mental health issues, or alcohol/drug related issues discovered during the performance of this contract must be reported to the USAG Security Office.

3.9. Search and Seizure:
Contractor personnel and property shall be subject to search and seizure upon entering the confines of Fort Meade and upon leaving the confines of Fort Meade.

3.10. Environmental Protection:
The Contractor shall manage all waste generated by any work performed under this contract in accordance with all applicable Federal (Clean Air Act; Toxic Substances Control Act; Solid Waste Disposal Act; Resource Conservation Recovery Act; Clean Water Act; Safe Drinking Water Act) and State laws and regulations; Executive Orders, or other types of rulings having the effect of the law. In case of a conflict among these laws and regulations, the most stringent law or regulation shall apply.

3.10.1. Environmental Management System:
Fort Meade has implemented an Environmental Management System (EMS) to address the environmental impacts of its activities. All Contractors performing work at Fort Meade must be aware of and understand Fort Meade's Environmental Policy. Contractors are responsible for ensuring environmental practices conform to Fort Meade's EMS.

3.10.2. Solid Waste and Recycling:
The Contractor shall recycle materials, such as, cardboard, paper, aluminum/metal cans, plastic, and glass jars/bottles, scrap metal, concrete, asphalt, and manage solid waste generated during the project in accordance with Federal and State laws and regulations. The Contractor shall provide the Government with records of the type, weight, and disposal/recycling location for all recyclable materials and solid waste sent offsite for recycling and/or disposal.

3.10.3. Hazardous Waste:
The Contractor shall manage hazardous waste in accordance with Federal and State laws and regulations. The contractor may not sign waste profiles or manifests on behalf of Fort Meade. The contractor must follow Fort Meade's Manifest SOP.

3.10.4. Hazardous Material:
The Contractor shall furnish and store all hazardous materials in a storage container or other secondary containment system. Maintenance of on-site equipment is not authorized.

3.10.5. Storm Water Pollution Prevention:
The Contractor shall implement measures to prevent storm water pollution in accordance with Federal and State laws and regulations. The Contractor shall furnish, install, maintain and operate erosion and sediment control measures, such as silt fences, storm drain protection, soil retention blankets, etc., prior to the start of any work that could cause pollution to state waters. The Contractor shall keep paved roads, parking lots and walkways clear of dirt, debris, and other materials. The Contractor shall remove immediately any dirt, debris, or materials deposited onto these surfaces. The Contractor shall furnish and use drip pans under equipment oil and fuel tanks, and cover equipment when not in use and during rain events.

It is the responsibility of the contractor to provide all design, approval and installation of storm water management requirements for the project. Regulatory storm water management requirements include construction erosion & sediment control and post-construction quantity and quality storm water runoff management. If project disturbance exceeds 5,000 square feet (SF) or more than 100 cubic yards, then an erosion and sediment control permit must be approved by the Maryland Regulations (COMAR 26.17.01). If project disturbance exceeds 5,000 SF, then a storm water management permit must be approved by the MDE before groundbreaking can occur pursuant to (COMAR 26.17.02). Projects disturbing one acre or more must obtain a National

Pollutant Discharge Elimination System (NPDES) permit pursuant to the General Permit for Construction Activities. It is the responsibility of the contractor to maintain all erosion & sediment control devices during construction to ensure minimal impacts on the environment. The contractor must conduct all required inspections of erosion & sediment control devices to ensure they are maintained and properly functioning. Once constriction is complete, the contractor must provide the MDE with as-built drawings of all post-construction quantity and quality storm water management devices for compliance permit termination.

3.10.6. Air Quality:
The Contractor shall take all necessary reasonable control measures to reduce air pollution by any material or equipment used to carry out the requirements of this contract. This includes, but is not limited to, wetting down dry materials as necessary to prevent blowing dust, and keeping waste and materials in covered containers.

3.11. Damage To Government Owned Property:
The Contractor shall be held liable for damage or destruction of Government property (e.g. buildings, vehicles, sidewalks, headstones, etc.) caused by the Contractor, Contractor personnel or representatives of the Contractor in the performance of services in this contract. The Contractor shall take corrective action in a reasonable amount of time to repair damages or replace destroyed property. If the Contractor fails to take the required corrective action, the Government retains the right to exercise other rights or remedies available to the Contracting Officer.

3.12. Damage To Contractor Owned Property:
The Contractor shall be held liable for damage or destruction of privately owned property (e.g. privately owned vehicles), as defined in this SOW, caused by the Contractor, Contractor personnel or representatives of the Contractor in the performance of services in this contract. The Contractor shall take corrective action in a reasonable amount of time to repair damages or replace destroyed property.

3.13. Damage Reports:
The Contractor shall submit a damage report to the Contracting Officer and designated representative within two (2) workdays of damage or destruction to Government property or privately owned property caused by the Contractor, Contractor personnel or representatives of the Contractor. The damage report shall detail the facts and the extent of damages or destruction, and the corrective action taken by the Contractor to repair damages or replace destroyed property.

4. Government Furnished Property, Equipment, & Services:

4.1. Utilities:
Government supplied water, electricity and other utilities will be available for this contract needs. The Contractor shall be responsible for operating under conditions that preclude the waste of utilities to accomplish work performed under this contract.

5. Contractor Furnished Property, Equipment, & Services:

5.1. General:
Except for items specifically identified as Government furnished in Part 4, the Contractor shall furnish all supplies, equipment, supervision, materials and services necessary to fulfil the requirements of this contract.

5.2. Permits, Taxes, Licenses, Ordinances, and Regulations:
The Contractor shall, at his own expense, obtain all necessary permits, give all notices, pay all license fees and applicable taxes, comply with municipal, state, and Federal laws, ordinances, rules and regulations applicable to the business carried on under this contract.

5.3. Equipment Staging and Security:
The Contractor shall take full responsibility for equipment stored or left onsite. The Government does not take responsibility for the security of Contractor's equipment or project material while the contract/project is being carried out. The Government is neither responsible nor liable for any of the Contractor's material, equipment stored either inside or outside the work area.

Attachment 1

REFERENCES

Project Specifications:
The whole building design guide - Unified Facilities Guide Specifications (UFGS)
Website https://www.wbdg.org/ccb/browse_cat.php?c=3

UFGS 00 01 15-----------List of Drawings
UFGS 01 11 00-----------Summary of Work
UFGS 01 14 00-----------Work Restrictions
UFGS 01 30 00-----------Administrative Requirements
UFGS 01 32 01-----------Project Schedule
UFGS 01 32 16-----------Progress Documentation
UFGS 01 33 00-----------Submittal Procedures
UFGS 01 35 26-----------Governmental Safety Requirements
UFGS 01 45 00----------Quality Control
UFGS 01 57 19-----------Temporary Environmental Controls
UFGS 01 78 00-----------Closeout Submittals
UFGS 01 78 23-----------Operation and Maintenance Data
UFGS 02 41 00-----------[Demolition] [and] [Deconstruction]

TECHNICAL EXHIBITS:

- Technical Exhibit 1: Emergency Contact Phone List.

Technical Exhibit 8: Emergency Phone Contact List

Important Contact Phone List:	**Fort Meade**	
PMO (Provost Marshall Office)	301-677-6600	
Fire Department	301-677-3162	
Safety Office	301-677-4231	
COR – Dipak J. Patel, P.E.	Office: 301-677-9445	

IN-CASE OF EMERGENCY CALL: 911